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Stainless Steel Grade 455 | UNS# S45500

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Stainless Steel Grade Custom 455 | UNS# S45500 is an advanced martensitic alloy included in the precipitation hardening stainless steels. Combining corrosion resistance, hardenability, ease of fabrication, and yield strength that is 3 times the strength of grade 304. Custom 455 is known for being soft and formable in the annealed condition, but single-step aging treatments result in an exceptionally high yield strength that also demonstrates good ductility and toughness. Applications generally include heat exchangers, conveyors, food processing and similar industries.

Stainless Steel Grade Custom 455 Specifications:

- [AMS 5617](#)
-



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Industries' locations:

- [Stainless Steel Grade Custom 455 Round Bar](#)

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Stainless Steel Custom 455 Applications:

- Heat Exchangers
- Surface Condensers
- Conveyors
- Food Processing

DISCLAIMER: Alloy data sheets are for reference only, not for design purposes.

CHEMICAL COMPOSITION				
Fe 73.0%	Cr 11.8%	Ni 8.50%	Nb 2.0%	Cu 2.0%
Ti 1.10%	Mn 0.50%	Mo 0.50%	Si 0.50%	C 0.50%
P 0.040%	S 0.030%			

PHYSICAL PROPERTIES					
Density	Electrical conductivity (%)	Thermal conductivity (BTU-in/hr-ft ² -°F)	Specific heat capacity	Thermal expansion (1/°F)	Melting Point (°F)

MECHANICAL PROPERTIES		
PROPERTY	MINIMUM	MAXIMUM
Yield Strength Range (Rp _{0.2} Offset) (ksi)	115	195
Tensile Strength Range (R _m) (ksi)	145	250
Elongation (AL=5 x Ø)	10.0%	14.0%
Brinell Hardness (H)	297	472
Modulus of Elasticity (ksi) (E): 29000		
Machineability General Index: 28%		